Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/10/2017
PI Name:	Deymier, Alix C. Ph.D.		
Project Title:	Effect of Unloading on the Structure and Mechanics of the Rotator Cuff Tendon-to-Bone Insertion		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	alix.c.deymier@gmail.com	**Fax:**	FY 212-342-6193
PI Organization Type:	UNIVERSITY	**Phone:**	212-305-7965
Organization Name:	Columbia University Medical Center		
PI Address 1:	Department of Orthopaedic Surgery		
PI Address 2:	William Black Bldg Rm 14-1408		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10032-3702	**Congressional District:**	13
Comments:	NOTE: Also known as Alix Deymier-Black; former affiliation Washington University School of Medicine (Ed., 3/8/17)		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 NSBRI-RFA-13-01 Postdoctoral Fellowships
Start Date:	11/01/2013	**End Date:**	10/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is now 10/31/2016 per NSBRI (Ed., 10/13/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thomopoulos, Stavros (MENTOR/ Washington University)		
Grant/Contract No.:	NCC 9-58-PF03503		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP The goal of this project is to investigate the effect of unloading on the tendon-to-bone attachment at a number of hierarchical scales. At the nanometer scale, I proposed to examine changes in the organization of the bone mineral with unloading via Transmission electron microscopy-electron energy loss spectroscopy (TEM-EELS). At a scale on the order of micrometers, I was interested in employing Raman and high-energy nano-x-ray diffraction (XRD) to identify modifications made to the mineral content, composition, structure, and organization in the interfacial tissue. Micro-mechanical testing would be employed to examine changes in the mechanics at the micron scale. At the millimeter scale, micro-computed tomography (μCT) and tensile testing were employed to determine the effects of unloading on bone quantity and quality and tissue mechanics. In the past three years, significant discoveries have been made explaining how the structure and mechanics of the tendon-to-bone attachment vary with unloading. Botulinum Toxin A (Btx) injections into the supraspinatus muscle were employed as an Earthbound model of disuse and unloading. This past year, I found that although there was no significant difference in the elastic properties of the attachment size with unloading, there were significant differences in the failure properties, both the strength and toughness. This modification in the failure properties appears to be two pronged: (1) a compromised complaint zone and (2) loss of enthesial bone. Micro-mechanical testing of the attachment site indicated that it contains a region of high deformation which may serve to dissipate stresses and minimize failure. However, the experimental assays performed here have shown that there are significant modifications to the compliant zone including a decrease in crystal size and carbonate content as well as an increase in mineral misalignment. Taken together, these effects will lead to stiffening of the energy absorbing compliance zone and an increase in fracture risk. In addition, unloaded samples have exhibited an increase in the bone loss nearing 40%. This loss of trabecular structure, which acts as a support for the attachment site, will increase bending at the attachment site and increase the tissue deformation and extent of failure. These results help to explain the measured increase in tissue removal during attachment failure of unloaded samples compared to loaded samples. In comparison, the elastic properties of the tendon-to-bone attachment, including the stiffness and resilience, remains unchanged indicating that the load bearing structures remain intact. However, the cross-sectional area of the tendon decreases with unloading while the attachment footprint increases. We propose that the change in cross-sectional area is due to changes in the non-load bearing tendon sheath. Combined with the decrease in crystal size and mineral carbonate content, these results indicate that unloading may induce a local acidic environment via glycolysis that dissolves tendon and bone structures. However, these have no effect on the tissue elastic properties. Finally, this study has provided some unique information about the organization of mineral across the attachment. At the micron scale, the mineral exhibits a gradient in size and orientation that is correlated to its location with respect to the collagen matrix. In both loaded and unloaded samples, the mineral crystals go from being uniquely extrafibrillar and free to grow in a disordered fashion in the regions of low mineralization to being intrafibrillar, constrained, and aligned in the regions of high mineralization. This shift in mineral organization directly affects percolation thresholds resulting in changes in the residual and root mean square strain as well as the local mechanical properties. These results indicate that then tendon-to-bone attachment uses a variety of structural techniques to finely control the local mechanical properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Rotator cuff tears are extremely prevalent, especially in the elderly population (~50% prevalence in individuals over 80 years). Even in the best of situations these tears are difficult to repair with a failure rate for repaired rotator cuffs as high as 94%. Rotator cuff tears tend to occur at the interface between tendon and bone. Such interfaces between dissimilar materials are prone to stress concentrations and increased failure risk. In healthy tissue, a number of structural mechanisms such as gradients in mineral content, collagen orientation, and matrix composition serve to dissipate these stress concentrations. The increased occurrence of rotator cuff injuries in the elderly population suggests that there may be changes in the interfacial structure due to unloading as a result of disuse or decreased use of the shoulder. Understanding how changes in the enthesis structure affect the mechanics of the insertion in loaded and unloaded systems will help us to develop enhanced techniques for treatment and repair. Therefore, the research performed in this project will not only help the astronaut population, but will also provide essential information in regards to the mechanics of rotator cuff tissues and how they respond to use and disuse.
Task Progress:	This project focuses on examining the mechanical and structural changes induced in the tendon-to-bone attachment during unloading at multiple length scales. This includes examining the tensile mechanics and bone structure at the millimeter scale, the micro-mechanics and mineral organization, composition, and structure at the micrometer scale, and the mineral organization relative to the collagen at the nano-meter scale. In all cases, I employed injections of Botulinum Toxin A (Btx) in the supraspinatus muscle, which induces local paralysis, as an analogue for unloading and microgravity. Mechanical testing indicated that there were significant changes to the failure properties but not elastic properties of the attachment. Strength and toughness are likely reduced due to (1) modifications of the compliant zone and (2) bone loss. The compliant zone, whose presence was confirmed during micro-mechanical testing, exhibits decreased crystal size, increased crystal misalignment, changes in mineral location relative to the collagen matrix, and decreased mineral carbonation with unloading. Using a series of models, I showed that these factors will stiffen the compliant zone decreasing attachment stiffness. In addition, the 40% trabecular bone loss will cause an increase in local deformation and explained the measured increase in tissue removal during failure. The cross-sectional area of the tendon was shown to decrease with unloading. Taken together with the decrease in crystal size and mineral carbonate content, this suggests that unloading may create a local acidic environment that dissolves disordered collagen and carbonated apatite crystals. This would be in keeping with the high levels of glycolysis previously seen in space flight and hind-limb unloaded tissues. Micrometer scale analysis of the attachment area showed that both loaded and unloaded samples exhibit gradients in mineral spanning approximately 10 μm. Along this gradient, the residual strain and root mean square strain become more compressive with increasing mineralization. This increase in compressive stress may indicate that crystals are becoming more confined and reaching a percolation threshold. The level of this threshold would be controlled by the structural differences in crystal shape, orientation, and location and precisely control the local mechanics. This indicates that the attachment uses a complex number of variables to control local mechanics. Overall, all of the proposed tasks have been achieved over the past 3 years. We now have a much deeper understanding of how the tendon-to-bone attachment reacts to unloading from both a structural and mechanical perspective.

Bibliography Type:	Description: (Last Updated: 10/19/2020)
Abstracts for Journals and Proceedings	Deymier-Black AC, Schwartz AG, Thomopoulos S. "Effect of Unloading on the Structure and Mechanics of the Rotator Cuff Tendon-to-Bone Insertion." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Deymier-Black AC, Nair AK, Depalle B, Quin Z, Arcot K, Yoder CH, Buehler MJ, Thomopoulos S, Genin GM, Pasteris JD. "What Determines the Size and Mechanics of Bone-like Mineral?" 62nd Annual Meeting of the Orthopaedic Research Society, Orlando, FL, March 5-8, 2016. 62nd Annual Meeting of the Orthopaedic Research Society, Orlando, FL, March 5-8, 2016. , Mar-2016
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Articles in Peer-reviewed Journals	Deymier AC, Schwartz AG, Cai Z, Daulton TL, Pasteris JD, Genin GM, Thomopoulos S. "The multiscale structural and mechanical effects of mouse supraspinatus muscle unloading on the mature enthesis." Acta Biomater. 2019 Jan 1;83:302-13. https://doi.org/10.1016/j.actbio.2018.10.024 ; PMID: 30342287; PMCID: PMC6343501 , Jan-2019
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Awards	Deymier A. "2nd Place NSBRI Dr. David Watson Post-Doctoral Fellow Poster Award, February 2016." Feb-2016